

Movie Explorer Group Project Report

INTRODUCTION

This project aims to create a console-based Movie Explorer application. Its main objectives are to help users search for movies and TV shows, view details, and quickly discover trending titles using real data from the TMDb API.

This report follows a clear structure: Introduction, Background, Specifications and Design, Implementation and Execution, Testing and Evaluation, and Conclusion.

BACKGROUND

Many people struggle to remember movies they want to watch or feel overwhelmed by the number of options available on streaming platforms. Our project provides an easy-to-use console application that allows users to search, explore, and make quick decisions about movies and TV shows.

Alongside solving this problem, the project allows our team to apply and practise Python programming, database handling, API integration, testing, and collaborative development.

SPECIFICATIONS AND DESIGN

Technical Requirements

- Python console application
- Local SQL database using SQLite (to store users, favourites, and history)
- API integration to retrieve movie and TV show information via TMDb
- Core logic for searching, recommendations, and managing favourites/history
- Modular code structure
- Error handling and input validation

Non-Technical Requirements

- User-friendly text-based menus
- Clear instructions for beginners
- Reliable and consistent behaviour
- Data privacy — only essential information stored

Design and Architecture

The system is designed in six main modules, plus tests:

1. Database & ORM Module (Laila)

- SQLite database schema for users, favourites, and history
- ORM or helper functions for reading/writing data

2. API Module (Stacey)

- Requests to TMDb API
- Parsing and returning structured movie/TV show information

3. Core Logic Module (Khadija)

- Processing searches and recommendations
- Managing favourites and history

4. Utils Module (Nada)

- Helper functions for formatting, validation, and console output
- Using Python libraries like itertools and collections

5. Main/Run Script (Zara)

- User interface and menu navigation
- Orchestrates calls to all other modules

6. Tests (Ayesha)

- Unit tests for all modules

High-level Architecture Flow

User Input → Validation → Database operations → Core Logic → API call → Output to user

Movie Explorer Architecture Diagram

